NISM SERIES XV · JANUARY 2026 (REVISED EDITION FOR THE EXAM EFFECTIVE 20 JANUARY 2026; EXACT VERSION LABEL NOT CONFIRMED ONLINE) WORKBOOK · STUDY NOTES

# Research Analyst

Read the economy, the industry and the company like a detective, then put a fair price on the story.

Imagine a food critic who tastes a restaurant's dishes, checks the kitchen and the owner's track record, and then tells readers whether the meal is worth the price. A Research Analyst does the same for shares, bonds and commodities: studies the economy, the industry and the company, values the security and publishes a Buy / Hold / Sell view. Because thousands of investors act on those calls, SEBI wants every analyst to know the craft (accounts, valuation, charts) and the rules (disclosures, conflicts, no trading against your own call). This NISM exam is the entry ticket that proves both.

## 📋 Exam pattern

| 100 | 120 min | 60% | 25% of the marks assigned to the question are deducted for every wrong answer |
|---|---|---|---|
| Total marks | Duration | Pass mark | Negative marking |

100 questions: 80 MCQs + 5 case-based question sets (4 questions each) · Revised exam effective 20 January 2026 (NISM circular dated 5 December 2025). Fee ₹1,500 incl. GST. Certificate is issued only if PAN is on record. Technical Analysis (15 marks), Financial Analysis (12), Valuation (12) and Legal & Regulatory (10) carry about half the paper. Expect numericals on ratios, valuation multiples, returns, Sharpe/Treynor/Jensen and corporate-action adjustments; keep a basic calculator mindset.

## 👥 Who needs it

- Individual research analysts registered with SEBI under the SEBI (Research Analysts) Regulations, 2014 (including part-time research analysts)
- Partners of a research analyst firm, the principal officer and all 'persons associated with research services' (employees who prepare reports, give recommendations or deal with clients)
- Analysts employed by research entities such as stock brokers, merchant bankers and investment banks that publish research in their own name
- Anyone who wants to give buy/sell/hold calls, price targets, model portfolios or trading calls for consideration (registration plus NISM certification is required to do so legally)

## 🧭 Exam strategy

- Budget about 1 minute per MCQ: do the 80 MCQs in ~70 minutes, leave ~40 minutes for the 5 cases (20 questions) and ~10 minutes to review.
- Read each case once fully, write down the key numbers (sales, EBITDA, debt, cash, shares, beta, Rf, Rm), then answer all four questions from that sheet.
- With 25% negative marking, guess only when you can eliminate at least two options; leave pure guesses blank.
- Secure the big four: Technical Analysis (15), Financial Analysis (12), Valuation (12) and Legal & Regulatory (10) - that is 49 marks.
- For numericals, check units (₹ crore vs per share) and whether the question wants EV or equity value - the distractors are usually those slips.
- For regulation questions, anchor to the exact numbers: 75 clients, ₹1,51,000, 30/5 days, 40/10 days, 1%, 12 months, 5 years, 21 days.
- In 'which is NOT' questions, test each option against the rule rather than hunting for the right-sounding one.
- For technical-analysis questions, first decide 'reversal or continuation' and 'bullish or bearish' - that eliminates most options.



Chapter 1 · Weight: 1 mark

# Introduction to Research Analyst Profession

Who a research analyst is, what the job involves day to day, how analysts should deal with companies and clients, and the qualities that make a good one. It sets the ethical tone for the rest of the book.

**MAKE IT STICK**

Think of a sports scout. He watches matches, studies stats, talks to coaches, but never takes a bribe from a player's agent, and his report must say clearly why he rates someone. A research analyst is the scout of the stock market.

1. Primary role: gather and analyse information on the economy, industries and companies, value securities and communicate investment views (Buy / Hold / Sell, price targets) through research reports.
2. Sell-side analysts work for brokers/investment banks and publish for clients; buy-side analysts work for mutual funds, insurers, PMS, hedge funds and advise their own fund managers; independent analysts sell research directly.
3. Core responsibilities: building financial models, tracking results and news, meeting managements, writing reports, updating views, and answering client queries.
4. Interaction with companies: rely on public information and legitimate analysis (the 'mosaic' approach); never seek or use unpublished price-sensitive information (UPSI).

5 Interaction with clients: give fair, balanced, well-reasoned views; disclose conflicts of interest; never guarantee returns.

6 Desired qualities: integrity, independence, curiosity, scepticism, numeracy, attention to detail, clear writing and the courage to give an unpopular rating.

7 A good analyst separates facts from opinions and states the assumptions behind every forecast.

8 Research must stay independent of investment-banking or brokerage business pressures (backed by SEBI RA Regulations on compensation and supervision).

Mosaic theory
**Combining public + non-material non-public bits into a conclusion is legitimate**

UPSI
**Never seek, use or pass on**

Analyst types
**Sell-side, buy-side, independent**

🎯 EXAM TIP
Expect one conceptual question: buy-side vs sell-side, or which behaviour with a company management is acceptable (public data, plant visits) vs unacceptable (asking for next quarter's numbers).

⚠️ TRAP
Mosaic analysis is NOT insider trading. Piecing together public information and non-material details is allowed; using even one piece of UPSI is not.



Chapter 2 · Weight: 2 marks

## Introduction to Securities Market

A tour of the Indian securities market: what counts as a security, the main instruments, how money is raised in the primary market and traded in the secondary market, who the participants are, and how derivatives and demat work.

MAKE IT STICK
The primary market is the car showroom where a brand-new car (share) is sold by the maker; the secondary market is the used-car bazaar where owners trade among themselves. The exchange is the bazaar's organiser and the clearing corporation is the escrow agent who makes sure money and keys change hands.

1 Securities include shares, bonds, debentures, derivatives, units of mutual funds and collective investment schemes, government securities and other instruments declared as securities (Securities Contracts (Regulation) Act, 1956).

2 Masala bonds are rupee-denominated bonds issued in overseas markets, so the currency risk sits with the foreign investor.

3 Warrants give the right to subscribe to shares at a pre-set price in the future; ETFs trade on the exchange like a share but hold a basket of securities.

4 Primary market routes: IPO (first public issue), FPO (further public offer), rights issue (to existing holders), bonus issue (free shares from reserves), private placement, preferential allotment, QIP (only to qualified institutional buyers), OFS (existing holders sell; company gets no money) and ESOPs.

5 Secondary market: OTC (bilateral) vs exchange-traded; the clearing corporation becomes the central counterparty and guarantees settlement; Indian equities settle on T+1.

6 Intermediaries: stock exchanges, clearing corporations, depositories (NSDL, CDSL), depository participants, brokers, authorised persons, custodians, merchant bankers, underwriters.

7 Institutional participants: FPIs, mutual funds, insurers, pension funds, AIFs, VC/PE funds; P-notes (ODIs) let unregistered overseas investors take exposure through registered FPIs.

8 Cash/spot trade settles immediately or T+1; a 'tom' trade settles tomorrow; forwards are customised OTC contracts; futures are standardised exchange contracts; options give a right but not an obligation; swaps exchange cash-flow streams.

9 Uses of derivatives: hedging (reduce risk), speculation/trading (take a view) and arbitrage (lock in a riskless price gap).

10 Pledging shares: promoters often pledge shares to raise loans; high pledge levels are a red flag because a price fall can trigger margin calls and forced selling.

11 Dematerialisation converts physical certificates into electronic holdings in a demat account; rematerialisation converts them back to paper.

12 Proxy advisory firms advise institutional investors on how to vote at shareholder meetings; they are regulated under the RA Regulations.

Equity settlement cycle
**T+1**

Depositories
**NSDL and CDSL**

QIP
**Placement only to QIBs**

OFS
**Money goes to selling shareholders, not the company**

Masala bond
**Rupee-denominated, issued abroad**

🎯 EXAM TIP
Who receives the money in an OFS vs IPO; difference between forward and futures; what a P-note is; dematerialisation vs rematerialisation.

⚠️ TRAP
In a bonus issue no money is raised at all, while in a rights issue the company raises fresh cash from existing shareholders. Do not call a bonus a 'primary market fund-raise'.

Chapter 3 · Weight: 2 marks

# Terminology in Equity and Debt Markets

The vocabulary analysts speak every day: different meanings of 'value', market cap and enterprise value, EPS and valuation ratios, bond jargon like coupon, YTM and duration, and commodity terms like basis, contango and backwardation.

**MAKE IT STICK**

A house has a printed price on the sale deed (face value), what the books say it is worth after depreciation (book value), what a buyer will pay today (market value), what it would cost to rebuild (replacement value) and what it is truly worth based on the rent it can earn (intrinsic value). Same house, five numbers.

1. Face value is the nominal value printed on the share; book value per share = net worth / shares; market value = traded price; replacement value = cost to recreate the assets; intrinsic value = value of expected future cash flows.
2. Market capitalisation = share price x number of shares outstanding.
3. Enterprise value (EV) = market cap + debt + preference capital + minority interest - cash and cash equivalents. EV is the value of the whole business to all capital providers.
4. Earnings can be historical (last reported year), trailing (last 12 months, TTM) or forward (estimates for next year).
5. EPS = net profit attributable to equity shareholders / weighted average shares; DPS = dividend per share.
6. P/E = price / EPS; P/S = market cap / sales; P/B = price / book value per share.
7. Differential voting rights (DVR) shares carry different (usually lower) voting rights than ordinary shares and often trade at a discount.
8. Bond terms: face value, coupon rate (interest on face value), maturity, principal, redemption; current yield = annual coupon / market price.
9. Yield to maturity (YTM) is the single discount rate that equates the bond's price to the present value of all its cash flows if held to maturity.
10. Bond prices and yields move inversely; duration measures the price sensitivity of a bond to interest-rate changes (higher duration = more sensitive).
11. Bond types: zero-coupon (issued at discount, no coupon), floating-rate (coupon reset to a benchmark), convertible, perpetual (no maturity), inflation-indexed, principal-protected notes, foreign-currency and external bonds.
12. Commodity terms: basis = spot - futures; contango = futures above spot (normal for storable commodities); backwardation = futures below spot (usually tight supply); cost of carry = storage + insurance + financing.

EV
**Mcap + Debt + Pref + Minority − Cash**

Current yield
**Annual coupon / Market price**

Contango
**Futures > Spot**

Backwardation
**Futures < Spot**

Basis
**Spot − Futures**

Zero-coupon bond duration
**Equals its maturity**

🎯 EXAM TIP
Quick numericals on market cap, EV, current yield and identifying contango vs backwardation. Also the inverse price-yield rule.

⚠️ TRAP
Cash is SUBTRACTED in EV, debt is ADDED. Students often add cash because 'cash is valuable' - but an acquirer gets the cash back, so it reduces the net cost.



Chapter 4 · Weight: 5 marks

## Fundamentals of Research

The menu of research approaches - active vs passive investing, technical, fundamental (top-down E-I-C and bottom-up), quantitative and behavioural - plus the basics of commodity research and how commodities affect equities.

MAKE IT STICK
To pick a mango you can look at its colour and how others are buying (technical), cut it and check the pulp (fundamental), run a sugar-content test on 1,000 mangoes (quantitative), or notice that you always overpay at the first stall you see (behavioural). Good analysts use all four lenses.

1. Passive investing tracks an index at low cost and accepts market returns; active investing tries to beat the benchmark through research - which is why analysts exist.
2. Research adds value by finding mispricing; legitimate research uses public information and the mosaic approach, never insider information.
3. Technical analysis studies price and volume history to identify trends and patterns; it assumes prices discount everything and history tends to repeat.
4. Fundamental analysis estimates intrinsic value from economy, industry and company factors (E-I-C framework).
5. Top-down: start with the economy, pick attractive industries, then select companies. Bottom-up: start with strong individual companies regardless of macro or sector view.
6. Quantitative (econometric) analysis uses statistical models, factor models and large datasets to identify relationships and screen securities.
7. Behavioural finance recognises that investors are not fully rational: biases such as overconfidence, herding, anchoring and loss aversion can push prices away from value.
8. Commodity research looks at supply, demand, seasonality, geopolitics, macroeconomics, government intervention and inventory levels.

9 Commodity prices feed into equities: e.g., a crude-oil rise hurts paints, airlines and tyre makers (input cost) but helps upstream oil producers.

10 Case studies and historical box items (bubbles and crashes) show how ignoring fundamentals or herd behaviour leads to losses.

E-I-C
**Economy → Industry → Company**

Top-down
**Macro first, stock last**

Bottom-up
**Stock first, macro is secondary**

Passive investing
**Index-tracking, low cost, no stock picking**

🎯 EXAM TIP
Match the approach to its description (technical vs fundamental vs quant vs behavioural), and top-down vs bottom-up sequencing.

⚠️ TRAP
Top-down is NOT 'start with the company and go up to the economy' - it starts at the top (economy). Bottom-up starts with the company.

Chapter 5 · Weight: 5 marks

## Economic Analysis

How the big economy shapes company profits: micro vs macro basics, GDP, savings and investment, inflation and interest rates, FDI/FPI flows, fiscal and monetary policy, trade and currency, and economic cycles - plus where to find the data.

MAKE IT STICK
A company is a boat; the economy is the tide. A great boat still struggles in a falling tide (recession) and an average boat floats up in a rising tide (boom). Economic analysis is checking the tide chart before you sail.

1 Microeconomics studies individual consumers, firms and markets (demand, supply, price, elasticity); macroeconomics studies the economy as a whole (growth, inflation, employment, policy).

2 National income: GDP measures the value of final goods and services produced within the country; it can be computed by production, income or expenditure methods. GDP = C + I + G + (X - M).

3 Savings fund investment; a higher domestic savings rate supports capital formation and growth.

4 Inflation is measured by CPI (consumer prices) and WPI (wholesale prices); RBI targets CPI inflation of 4% with a tolerance band of +/-2%.

5 Monetary policy (RBI's Monetary Policy Committee) uses the repo rate, CRR, SLR and open market operations; rate cuts generally support equity valuations and rate-sensitive sectors.

6 Fiscal policy is the government's taxation and spending; fiscal deficit = total expenditure - total receipts excluding borrowings. High deficits can crowd out private borrowing and push up yields.

7 FDI is long-term strategic investment (generally 10% or more of a company); FPI is portfolio investment (below 10%) and is more volatile ('hot money').

8 Current account deficit and trade deficit put pressure on the rupee; a weaker rupee helps exporters (IT, pharma) and hurts importers (oil marketing, electronics).

9 Economic trends: secular (long-term structural), cyclical (business cycle of expansion, peak, contraction, trough) and seasonal (calendar-linked, e.g., festive demand, monsoon).

10 Cyclical sectors (autos, metals, capital goods, real estate) swing with the cycle; defensive sectors (FMCG, pharma, utilities) are steadier.

11 Globalisation brings bigger markets, capital and technology but also contagion and import competition.

12 Sources: MoSPI/NSO (GDP, CPI, IIP), RBI (monetary data, Handbook of Statistics), Ministry of Finance (Budget, Economic Survey), Ministry of Commerce (trade data, WPI), IMF/World Bank.

| | | |
|---|---|---|
| GDP (expenditure)<br>**C + I + G + (X − M)** | RBI inflation target<br>**CPI 4% (band 2%–6%)** | FDI vs FPI threshold<br>**10% of the company's equity** |
| Business cycle<br>**Expansion → Peak → Contraction → Trough** | Weaker rupee helps<br>**Exporters (IT, pharma)** | |

🎯 EXAM TIP
Impact questions: 'If RBI cuts the repo rate, which sector benefits most?' or 'A depreciating rupee hurts which company?'. Also the type of trend (secular/cyclical/seasonal).

⚠️ TRAP
Fiscal policy = government (budget, taxes, spending). Monetary policy = RBI (rates, liquidity). Swapping these is the classic slip.



Chapter 6 · Weight: 8 marks

## Industry Analysis

Why the neighbourhood matters as much as the house: defining an industry, cyclicality, market sizing, life cycle, strategic frameworks (Porter, PESTLE, BCG, SCP), industry-specific KPIs, regulation, taxation and data sources.

MAKE IT STICK
The best swimmer in a drying pond still can't win. A brilliant telecom company in a price-war era earned little, while an ordinary company in a booming sector (say, early-stage digital payments) rode the wave. Industry analysis tells you whether the pond is filling or drying.

1 An industry is a group of companies producing similar products or services that compete for the same customers (e.g., cement, IT services, private banks).

2 Cyclical industries (cement, steel, autos) move with the economy; defensive industries (FMCG, pharma) are less affected; growth industries grow faster than GDP regardless.

3 Market sizing: TAM (total addressable market) estimated top-down (population x penetration x price) or bottom-up (units x price); trend analysis tracks growth, penetration and share.

4 Industry life cycle: pioneering/introduction → growth → maturity/consolidation → decline; valuations and competition differ at each stage.

5 Value migration: profits shift from outdated business models to new ones (e.g., film cameras to smartphones); secular trends drive this shift.

6 Porter's Five Forces: threat of new entrants, bargaining power of suppliers, bargaining power of buyers, threat of substitutes, and rivalry among existing competitors.

7 PESTLE: Political, Economic, Social, Technological, Legal and Environmental factors.

8 BCG matrix (market growth vs relative market share): Stars (high/high), Cash Cows (low growth, high share), Question Marks (high growth, low share), Dogs (low/low).

9 Structure-Conduct-Performance (SCP): market structure (concentration, entry barriers) shapes firm conduct (pricing, R&D) which determines performance (profitability).

10 Industry KPIs: units of pricing (₹/tonne for cement, ARPU for telecom, ₹/sq ft for real estate, per-kWh for power), constraints (capacity, raw materials, licences) and KPIs like NIM and GNPA for banks, same-store sales for retail, occupancy and ARR for hotels, capacity utilisation for cement.

11 Regulation can make or break an industry (tariff orders, price controls on drugs, licensing, environmental norms) - analysts must track it continuously.

12 Taxation: direct taxes (income tax, corporate tax), indirect taxes (GST, customs), securities taxes (STT, capital gains). Dividend Distribution Tax was abolished from 1 April 2020; dividends are now taxed in the hands of shareholders.

13 Industry information sources: industry associations (SIAM for autos, etc.), regulators (TRAI, IRDAI, RBI), government ministries, annual reports of peers, trade publications and research databases.

Porter's forces
**5: entrants, suppliers, buyers, substitutes, rivalry**

PESTLE
**6 factors**

Cash Cow
**Low growth, high market share**

Question Mark
**High growth, low market share**

Life cycle
**Pioneering → Growth → Maturity → Decline**

DDT
**Abolished from 1 April 2020**

🎯 EXAM TIP

Framework identification (which Porter force / BCG quadrant / PESTLE letter), matching an industry to its KPI or pricing unit, and life-cycle stage traits.

⚠️ TRAP

BCG axes are market GROWTH and RELATIVE MARKET SHARE. A Cash Cow has high share in a SLOW-growing market - not a fast one.

Chapter 7 · Weight: 6 marks

## Company Analysis – Business and Governance

The qualitative side of company research: understanding the business model, pricing power, competitive moat, SWOT, management quality, corporate governance, promoter holding, business risks, credit-rating history, ESG and information sources.

MAKE IT STICK

Two tea stalls on the same street sell the same chai. One has a loyal crowd because the owner remembers your name, buys milk cheaper in bulk and never waters it down. That invisible edge - loyalty, cost and honesty - is what this chapter teaches you to spot.

1. Understand the business model: what the company sells, to whom, how it earns money, its cost structure and capital needs.
2. Pricing power: the ability to raise prices without losing customers - a hallmark of strong brands, monopolies or essential products.
3. Competitive advantage (moat): brand, cost leadership, network effects, switching costs, patents/licences, distribution reach; ask if it is sustainable.
4. SWOT: Strengths and Weaknesses are internal; Opportunities and Threats are external.
5. Management quality: track record, capital allocation, execution, transparency, consistency between guidance and delivery.
6. Corporate governance: board independence, quality of independent directors, related-party transactions, auditor quality, minority-shareholder treatment, disclosure standards.
7. Promoter holding: high and stable holding signals commitment; rising pledge or steady selling is a warning sign.
8. Business risks: demand, competition, regulatory, technology obsolescence, concentration (customer/supplier), forex and commodity price risk.
9. Credit rating history: a series of downgrades signals deteriorating finances; a rating 'watch with negative implications' is an early warning.
10. ESG analysis looks at Environmental (emissions, waste), Social (labour, safety, community) and Governance (board, ethics) factors; top listed companies publish a Business Responsibility and Sustainability Report (BRSR).

11 Sources: annual reports, quarterly results, investor presentations, earnings-call transcripts, exchange filings, credit-rating rationales, MCA filings, management meetings and channel checks.

| SWOT internal | SWOT external | ESG report (India) | Pledge red flag |
|---|---|---|---|
| **Strengths, Weaknesses** | **Opportunities, Threats** | **BRSR** | **Rising promoter pledge** |

🎯 EXAM TIP
Classify an item as S/W/O/T, identify governance red flags (frequent auditor resignation, large related-party transactions, rising pledge), and moat types.

⚠️ TRAP
A threat like a new competitor or regulation is EXTERNAL - it belongs in 'T', not 'W', even though it hurts the company.



Chapter 8 · Weight: 12 marks

## Company Analysis – Financial Analysis

The numbers chapter: reading the balance sheet, P&L and cash-flow statement, notes to accounts and audit report, then using ratios (profitability, return, leverage, liquidity, efficiency), DuPont and peer comparison to judge quality and trajectory.

MAKE IT STICK
A doctor reads your blood report before prescribing. Ratios are the blood report of a company: high sugar (debt), low haemoglobin (cash flow), cholesterol (receivables piling up). One reading means little; the trend and comparison with healthy peers tells the story.

1 Standalone statements cover only the company; consolidated statements include subsidiaries (and share of associates/JVs) - analysts generally prefer consolidated for groups.

2 Balance sheet: assets = equity + liabilities; key lines are share capital, reserves, borrowings, trade payables, fixed assets, CWIP, investments, inventory, receivables and cash.

3 P&L: revenue → EBITDA (operating profit before depreciation) → EBIT → PBT → PAT; watch other income and exceptional items separately from core operations.

4 Cash-flow statement: operating (CFO), investing (CFI) and financing (CFF). Persistent profits with weak CFO is a quality red flag.

5 Free cash flow to firm (FCFF) ≈ CFO - capex (on an unlevered basis); free cash flow to equity (FCFE) = FCFF - interest x (1 - t) + net borrowing.

6 Profitability ratios: gross, EBITDA, EBIT and net margins (as % of revenue).

7 Return ratios: ROE = PAT / average shareholders' equity; ROCE = EBIT / capital employed (equity + debt); ROA = PAT / total assets.

8 Leverage: debt-equity = total debt / equity; interest coverage = EBIT / interest; net debt/EBITDA.

9 Liquidity: current ratio = current assets / current liabilities; quick ratio = (current assets - inventory) / current liabilities.

10 Efficiency: asset turnover = sales / total assets; inventory days = inventory / COGS x 365; receivable days = receivables / sales x 365; payable days = payables / purchases (or COGS) x 365; cash conversion cycle = inventory days + receivable days - payable days.

11 DuPont (3-step): ROE = net margin x asset turnover x equity multiplier (assets/equity). The 5-step version splits margin into tax burden, interest burden and EBIT margin.

12 Equity dilution: track how share count grew (QIPs, preferential issues, ESOPs, conversions); EPS growth matters more than profit growth for shareholders.

13 Notes to accounts reveal contingent liabilities, related-party transactions, accounting policy changes, segment data and pledges.

14 Audit report: a qualified opinion, adverse opinion, disclaimer or 'emphasis of matter', and frequent auditor changes, are signals of accounting risk.

15 Past performance is a guide, not a guarantee; use history cautiously and compare with peers on the same metrics and accounting basis.

ROE (DuPont)
**Net margin x Asset turnover x Equity multiplier**

ROCE
**EBIT / (Equity + Debt)**

Quick ratio
**(CA – Inventory) / CL**

Interest coverage
**EBIT / Interest**

Cash conversion cycle
**Inventory days + Receivable days – Payable days**

Red flag
**Profit rising, operating cash flow falling**

🎯 EXAM TIP
At least 3–5 numericals: ROE, ROCE, current/quick ratio, D/E, interest cover, working-capital days and DuPont decomposition. Also qualitative: which audit opinion is worst, what consolidated statements include.

⚠️ TRAP
ROE uses PAT and EQUITY; ROCE uses EBIT (pre-interest, pre-tax) and CAPITAL EMPLOYED (equity + debt). Mixing numerators and denominators is the commonest error.

Chapter 9 · Weight: 5 marks

# Corporate Actions

Events that change a company's share count, capital structure or shape: dividends, rights, bonus, splits, consolidation, mergers, demergers, schemes of arrangement, loan restructuring, buybacks, delisting/relisting and share swaps - and how to adjust per-share data for them.

**MAKE IT STICK**

Cutting a pizza into 8 slices instead of 4 (split) or getting a free slice for every slice you own (bonus) does not give you more pizza. But if the shop buys back slices from others (buyback) your share of the pizza grows. Corporate actions are mostly about who owns which slice.

1. Rationale: return cash (dividend, buyback), raise capital (rights), improve liquidity/affordability (split, bonus), restructure (merger, demerger) or exit the market (delisting).
2. Dividends: interim (declared by board) and final (approved by shareholders at AGM); key dates are record date and ex-date. Share price typically falls by about the dividend on the ex-date.
3. Rights issue: existing shareholders may buy new shares in proportion, usually at a discount; rights entitlements are tradable (renounceable). Theoretical ex-rights price (TERP) = (old shares x market price + rights shares x issue price) / total shares.
4. Bonus issue: free shares by capitalising reserves; share count rises, price adjusts down, total value and net worth unchanged (reserves move into share capital).
5. Stock split: face value is reduced (e.g., ₹10 to ₹2 gives 5 shares for 1); share capital and reserves are unchanged. Consolidation (reverse split) does the opposite.
6. For bonus and splits, historical EPS, DPS and prices must be adjusted for the new share count so comparisons stay meaningful.
7. Mergers and acquisitions: combine businesses for synergy, scale or market access; share swap ratio decides how many acquirer shares target holders receive.
8. Demerger / spin-off: a division is separated into a new listed company; shareholders get shares in the new entity, often unlocking value hidden in a conglomerate.
9. Schemes of arrangement (compromise or arrangement under the Companies Act) need approval of shareholders/creditors and the NCLT.
10. Loan restructuring changes debt terms (tenor, rate, conversion to equity) for stressed borrowers - analysts treat it as a signal of financial distress.
11. Buyback: company repurchases its own shares (tender offer or open market), reducing share count and usually raising EPS; it signals management thinks the stock is undervalued and returns surplus cash.
12. Delisting: voluntary (promoter acquires shares through a reverse book-building / price discovery process) or compulsory (by the exchange as a penalty); relisting is restricted for a cooling-off period afterwards.

13 Share swap: acquisition paid with the acquirer's shares instead of cash; target shareholders become shareholders of the acquirer.

| TERP | Bonus 1:1 | Split ₹10 → ₹1 |
|---|---|---|
| **(N x P + R x S) / (N + R)** | **Shares double, price ~halves, value unchanged** | **10 shares for 1** |

| Bonus vs split | Buybank |
|---|---|
| **Bonus capitalises reserves; split changes face value only** | **Fewer shares → higher EPS** |

🎯 EXAM TIP

Numericals on post-bonus/split share count and price, TERP for rights, adjusted EPS, and a merger swap-ratio calculation. Conceptual: which action changes face value, which affects reserves.

⚠️ TRAP

A bonus issue does NOT change face value but reduces reserves (moved to share capital); a split CHANGES face value but leaves reserves untouched. Neither creates value.



Chapter 10 · Weight: 12 marks

## Valuation Principles

Turning analysis into a number: price vs value, why we value, earnings- and asset-based sources of value, DCF, relative valuation with earnings and asset multiples, transaction multiples, SOTP, new-age business metrics, CAPM, and how to keep valuation objective.

MAKE IT STICK

Price is what the vegetable vendor asks; value is what the vegetables are actually worth to your kitchen. Warren Buffett's line - 'price is what you pay, value is what you get' - is the whole chapter in one sentence.

1 Price is the market quote; value is an estimate of worth based on future cash flows or assets. Investment opportunities exist when price and value diverge.

2 Reasons to value: investing decisions, M&A, fund raising, restructuring, regulatory/tax requirements, dispute resolution.

3 Sources of value: earnings/cash-flow power (going concerns) and assets (asset-heavy, holding or liquidation situations).

4 Approaches: absolute (DCF, dividend discount), relative (multiples vs peers or history) and asset-based (NAV, replacement or liquidation value).

5 DCF: forecast free cash flows for an explicit period, add a terminal value, and discount at WACC (for FCFF → enterprise value) or cost of equity (for FCFE → equity value). Equity value = EV - net debt.

6 Terminal value (Gordon growth) = FCF(n+1) / (r - g); it usually forms a large share of DCF value, so it is very sensitive to r and g (g must be below r).

7 Dividend discount model (constant growth): value = D1 / (ke - g).

8 Earnings multiples: P/E, earnings yield (E/P, inverse of P/E), dividend yield (DPS/price), PEG = P/E ÷ EPS growth rate (%), EV/EBITDA, EV/EBIT and EV/Sales.

9 EV multiples are capital-structure neutral and useful for comparing firms with different debt levels; P/E is affected by leverage and accounting.

10 Asset multiples: P/B (banks, NBFCs), EV/capital employed, NAV for holding companies and real-estate firms; replacement cost for capital-intensive sectors.

11 Trading multiples come from listed peers; transaction multiples come from past M&A deals and include a control premium.

12 Sum-of-the-parts (SOTP): value each segment/subsidiary separately (often with different methods), add them and usually apply a holding-company discount for listed stakes.

13 New-age / loss-making businesses: EV/Sales, EV/GMV, price per user, customer lifetime value vs acquisition cost (LTV/CAC), unit economics and path to profitability.

14 CAPM: cost of equity = Rf + β x (Rm - Rf). Beta measures systematic (market) risk.

15 Objectivity: use multiple methods, cross-check assumptions, run sensitivity analysis, avoid anchoring on the current price, and disclose key assumptions.

| CAPM | Gordon growth | PEG | Earnings yield |
|---|---|---|---|
| **Ke = Rf + β (Rm – Rf)** | **Value = D1 / (ke – g)** | **P/E ÷ EPS growth (%)** | **EPS / Price = 1 / (P/E)** |

| Equity value from EV | FCFF discount rate |
|---|---|
| **EV – Debt + Cash** | **WACC; FCFE discount rate = Ke** |

🎯 EXAM TIP

Heavy numericals: DCF/terminal value, DDM, CAPM, PEG, EV/EBITDA → implied price per share, SOTP with holding discount. Conceptual: which multiple suits which sector, trading vs transaction multiples.

⚠️ TRAP

Match cash flow to discount rate: FCFF with WACC gives ENTERPRISE value; FCFE with cost of equity gives EQUITY value. Also, after EV/EBITDA you must subtract net debt before dividing by shares.

Chapter 11 · Weight: 5 marks

## Fundamental Analysis of Commodities

What moves commodity prices: supply-demand balance, big producers and consumers, the US dollar, international-domestic price links, crop and weather reports, inventory data, macro indicators, government policy and geopolitics - and how hedging with futures works.

**MAKE IT STICK**

Onion prices in your local market jump when unseasonal rain hits Maharashtra's crop, fall when the government allows imports, and spike again if exports are freed. A commodity analyst reads the sky, the warehouse and the government gazette, not a balance sheet.

1. Prices are set by the supply-demand balance; supply depends on production, carry-over stocks and imports, demand on consumption, industrial use and exports.
2. Major producers and consumers matter: e.g., OPEC+ for crude, China as the biggest consumer of base metals, India as a major consumer of gold and edible oil importer.
3. Most global commodities are priced in US dollars, so a stronger dollar (higher Dollar Index) generally pushes dollar commodity prices down; for Indian buyers, a weaker rupee raises the local price.
4. Domestic prices of internationally traded commodities track global prices adjusted for exchange rate, import duties and freight (import parity).
5. Agricultural commodities depend on crop reports (sowing, yield estimates), monsoon and weather forecasts (IMD), and global reports like USDA's WASDE.
6. Inventory, production and consumption data (e.g., crude oil inventories, LME metal stocks) signal tightness; falling inventories are bullish.
7. Macro indicators: GDP growth, industrial production, PMI, interest rates and inflation affect commodity demand; gold often rises with uncertainty and falling real rates.
8. Government policy: MSP, export bans, import duties, stock limits, buffer stock operations; geopolitics: wars, sanctions, supply disruptions.
9. Hedging locks in a price to reduce risk: a producer sells futures (short hedge), a consumer buys futures (long hedge).
10. Minimum-variance hedge ratio = correlation x (σ spot / σ futures); number of contracts = hedge ratio x exposure / contract size.
11. Hedging advantages: price certainty, easier planning and financing; disadvantages: gives up favourable price moves, basis risk, margin cash-flow needs.

| | |
|---|---|
| Hedge ratio | **ρ x σS / σF** |
| Producer hedge | **Short futures** |
| Consumer hedge | **Long futures** |
| Strong US dollar | **Usually bearish for dollar commodity prices** |
| Falling inventories | **Bullish signal** |

**🎯 EXAM TIP**

Hedge ratio and contract-count numericals, who goes long vs short, and the direction of impact from dollar, monsoon, export ban or inventory data.

**⚠️ TRAP**

A FARMER (who will sell the crop) hedges by SELLING futures; a flour MILL (who will buy wheat) hedges by BUYING futures. Hedge in the direction opposite to your natural exposure.

Chapter 12 · Weight: 7 marks

# Fundamentals of Risk and Return

Measuring returns (simple, annualised, compounded), the many kinds of investment risk, standard deviation and beta, sensitivity analysis, margin of safety, equity vs bond returns, risk-adjusted ratios (Sharpe, Treynor, Jensen), behavioural biases, investing wisdom and liquidity measures.

**MAKE IT STICK**

Two drivers reach Mysuru from Bengaluru in 3 hours. One drove steadily; the other overtook dangerously and braked hard. Same return, very different risk. Sharpe and Treynor ratios tell you who was the better driver per unit of risk taken.

1. Holding-period return = (ending value - beginning value + income) / beginning value.
2. Simple annualised return = HPR x (365 / days held); CAGR = (ending / beginning)^(1/years) - 1 (accounts for compounding).
3. Risks: inflation (purchasing power), interest-rate, business, market (systematic), credit/default, liquidity, call (bond called when rates fall), reinvestment, political and country risk.
4. Total risk = systematic (market, non-diversifiable, measured by beta) + unsystematic (company-specific, diversifiable).
5. Standard deviation measures total volatility of returns; beta = covariance(stock, market) / variance(market) measures sensitivity to the market. Market beta = 1.
6. Sensitivity analysis changes one input at a time (e.g., growth, margin, discount rate) to see how value changes; scenario analysis changes several together.
7. Margin of safety (Benjamin Graham): buy only when price is well below estimated intrinsic value; margin of safety = (intrinsic value - price) / intrinsic value.
8. Equities have historically given higher long-term returns than bonds but with higher volatility; the gap is the equity risk premium.
9. Sharpe ratio = (Rp - Rf) / σp (return per unit of total risk); Treynor ratio = (Rp - Rf) / βp (per unit of systematic risk); Jensen's alpha = Rp - [Rf + βp (Rm - Rf)] (excess over CAPM).
10. Behavioural biases: loss aversion, confirmation bias, ownership/endowment bias, gambler's fallacy, winner's curse, herd mentality, anchoring, projection bias.
11. Investment wisdom from greats (Graham, Buffett, Lynch and others): invest within your circle of competence, think long-term, insist on a margin of safety, be fearful when others are greedy.
12. Liquidity of shares is measured by traded volume/value, turnover ratio, bid-ask spread, impact cost and free float; illiquid stocks deserve a discount.

| Sharpe | Treynor | Jensen's alpha | CAGR |
|---|---|---|---|
| **(Rp – Rf) / σp** | **(Rp – Rf) / β** | **Rp – [Rf + β(Rm – Rf)]** | **(End/Begin)^(1/n) – 1** |

Diversifiable risk
**Unsystematic (company-specific)**

Margin of safety
**(IV − Price) / IV**

🎯 EXAM TIP
Numericals on HPR, CAGR, Sharpe, Treynor, Jensen; identify the bias from a short description; classify risks as systematic vs unsystematic.

⚠️ TRAP
Sharpe divides by STANDARD DEVIATION (total risk); Treynor divides by BETA (systematic risk). For a well-diversified portfolio Treynor is appropriate; for a single concentrated portfolio use Sharpe.



Chapter 13 · Weight: 5 marks

## Qualities of a Good Research Report

What makes a research report useful and compliant: clarity, accuracy, a clear recommendation with defined rating and time horizon, supporting data, disclosures, and the discipline of a checklist-based approach.

MAKE IT STICK
A pilot, however experienced, runs a pre-flight checklist every single time. A checklist-driven research report does the same: it stops a brilliant analyst from forgetting the boring item - like the related-party note or the disclosure - that brings the plane down.

1. A good report is clear, concise, accurate, timely, objective, logically structured and actionable; the investment thesis and recommendation appear upfront.
2. Facts must be based on reliable information; assumptions and estimates are stated clearly; recommendations must be corroborated by relevant data and analysis.
3. Rating conventions (e.g., Buy / Accumulate / Hold / Reduce / Sell) must be defined, including the time horizon and benchmark, and applied consistently.
4. Price target, valuation method and key risks to the call should be disclosed.
5. Where a rating or price target has been assigned for at least one year, the report shows a daily closing-price graph for the period or three years, whichever is shorter (per the RA Regulations).
6. Mandatory disclosures cover the analyst's and entity's financial interests (including holding 1% or more), compensation received from the subject company in the past 12 months, and other conflicts.
7. A checklist-based approach brings consistency, reduces omissions and bias, makes reports comparable across analysts and easier to review by compliance.
8. A sample checklist covers business understanding, industry, management and governance, financials, valuation, risks, and compliance/disclosure items.

Price chart in report
**If rating held ≥1 year: daily closes for the period or 3 years, whichever is shorter**

Ownership disclosure
**1% or more of subject company**

Compensation look-back
**Past 12 months**

Ratings
**Must define meaning, horizon and benchmark**

🎯 EXAM TIP
Which element is NOT essential/allowed in a report, the benefits of a checklist, and the specific disclosure thresholds (1%, 12 months, 3-year chart).

⚠️ TRAP
A long report is not a good report. Examiners like 'which of the following is a quality of a good report' with 'length' or 'guaranteed returns' as distractors.



Chapter 14 · Weight: 10 marks

## Legal and Regulatory Environment

The rulebook: who regulates what in Indian finance, the key laws (SCRA, SEBI Act, insider trading, PFUTP, RA Regulations, IBC), the RA code of conduct, conflict-of-interest rules and disclosures, and exchange surveillance measures ASM and GSM - including the 2024–25 amendments to the RA Regulations.

MAKE IT STICK
A referee doesn't play the match, but every player must know the rules or get a red card. SEBI is the referee; the RA Regulations are the rulebook; ASM/GSM are the yellow cards exchanges flash on overheated stocks.

1. Regulators: Ministry of Finance (overall policy), MCA (Companies Act, IBBI oversight), RBI (banks, NBFCs, money and forex markets), SEBI (securities markets), IRDAI (insurance), PFRDA (pension/NPS), IBBI (insolvency professionals) and WDRA (warehouses for commodity deliveries).

2. SCRA 1956 regulates stock exchanges and securities contracts; SEBI Act 1992 created SEBI as a statutory regulator to protect investors and develop and regulate the market.

3. SEBI (Prohibition of Insider Trading) Regulations 2015: no trading while in possession of UPSI; no communication of UPSI except for legitimate purposes; trading window closure and pre-clearance for designated persons.

4. SEBI (PFUTP) Regulations 2003 prohibit fraud, market manipulation, front-running, circular trading and misleading statements to induce trading.

5. RA Regulations (2014, amended Dec 2024 and 2025): no one may act as or hold out as a research analyst without SEBI registration; investment advisers, credit rating agencies and AMCs issuing research are exempt from registration but follow the conduct chapter.

6. Qualification (as amended 2025): graduate degree in any discipline (or CFA Charter) plus the NISM certification, or the NISM PG Programme in Securities Markets (Research Analysis); fresh NISM certification must be obtained before the existing one expires.

7 Part-time research analysts (individuals/partnership firms with other business or employment) may register; they can have at most 75 clients at any time and must call themselves 'part-time research analyst'.

8 Net-worth requirements were replaced by a deposit with a scheduled bank, marked as lien in favour of the supervisory body (RAASB): ₹1 lakh up to 150 clients, ₹2 lakh for 151–300, ₹5 lakh for 301–1,000 and ₹10 lakh above 1,000 clients.

9 BSE Ltd was recognised by SEBI (2024) as the Research Analyst Administration and Supervisory Body (RAASB); RAs must enlist with it.

10 Fees to individual and HUF clients (not accredited investors) are capped at ₹1,51,000 per family per year; advance fees can be collected for up to one year if the client agrees (relaxed in April 2025 from one quarter); fees are not taken in cash.

11 Personal trading: the RA and associates must not deal in a covered security 30 days before and 5 days after publishing a report, and must never trade against their own recommendation.

12 Quiet periods for research entities acting as manager/co-manager: 40 days after IPO pricing and 10 days after FPO pricing; 15 days before/after expiry of a lock-up agreement.

13 Disclosures: ownership of 1% or more of the subject company (end of previous month), compensation from the subject company in the past 12 months, any directorship/officer role, market-making, and the extent of AI-tool use.

14 Records kept for at least 5 years; annual compliance audit by an ICAI/ICSI/ICMAI member; client grievances to be resolved within 21 calendar days; research and distribution must be segregated at client level for the RA's family/group.

15 Code of Conduct (Third Schedule): honesty and good faith, diligence, conflict of interest, insider trading/front running, confidentiality, professional standard, compliance, responsibility of senior management.

16 Exchange surveillance: ASM (Additional Surveillance Measure) for unusual price/volume moves - e.g., higher margins; GSM (Graded Surveillance Measure) for stocks whose price does not match financial health - graded stages with restrictions like trade-for-trade and surveillance deposits. These are surveillance tools, not adverse judgements on the company.

| | |
|---|---|
| Part-time RA client cap | **75 clients at any time** |
| Fee cap (individual/HUF) | **₹1,51,000 per family per year** |
| Advance fee | **Up to 1 year, if client agrees** |
| Deposit ≤150 clients | **₹1 lakh (₹2L / ₹5L / ₹10L for bigger tiers)** |
| Trading blackout | **30 days before, 5 days after report** |
| IPO quiet period (manager) | **40 days after pricing** |
| FPO quiet period (manager) | **10 days after pricing** |
| Ownership disclosure | **≥1% of subject company** |
| Record retention | **At least 5 years** |
| Grievance redressal | **21 calendar days** |
| RAASB | **BSE Ltd** |

🎯 EXAM TIP

Expect ~10 questions: registration exemption, part-time RA cap, the 30/5-day blackout, quiet periods, 1%/12-month disclosures, regulator matching, PIT and PFUTP basics, ASM vs GSM, and code-of-conduct heads.

⚠️ TRAP

Investment advisers, credit rating agencies and AMCs issuing research are EXEMPT from RA registration but NOT from the conduct rules. Also: 30 days BEFORE and 5 days AFTER - not the other way round.



Chapter 15 · Weight: 15 marks

## Technical Analysis

Reading price charts: the philosophy of technical analysis, chart types, Dow Theory, trends, reversal and consolidation patterns, support and resistance, trendlines and channels, and indicators like moving averages, RSI, MACD and Bollinger Bands. It is the single heaviest chapter.

MAKE IT STICK

Footprints in wet sand tell you which way the crowd walked, how fast, and where they paused. Technical analysts read footprints of price and volume to guess where the crowd will walk next.

1. Premises: market action discounts everything, prices move in trends, and history (crowd psychology) tends to repeat.
2. Charts: line (closing prices), bar (OHLC), candlestick (OHLC with a coloured body), point & figure (price only, ignores time) and volume bars.
3. Dow Theory: averages discount everything; three trends - primary (a year or more), secondary (corrections lasting about 3 weeks to 3 months), minor (less than 3 weeks); primary trends have three phases (accumulation, public participation, distribution); averages must confirm each other; volume confirms the trend; a trend continues until a definite reversal signal.
4. Uptrend = higher highs and higher lows; downtrend = lower highs and lower lows; sideways = range-bound.
5. Reversal patterns: head and shoulders (top), inverse head and shoulders (bottom), double/triple tops and bottoms, rounding (saucer) tops/bottoms, V-reversals and key reversal days. A break of the neckline confirms H&S; target ≈ head-to-neckline distance projected from the break.
6. Consolidation (continuation) patterns: triangles (symmetrical, ascending, descending), flags, pennants, rectangles and wedges; breakout usually resumes the prior trend.
7. Support is a price level where buying stops a fall; resistance is where selling stops a rise; once broken, support often becomes resistance and vice versa (role reversal).
8. Trendlines join successive higher lows (uptrend) or lower highs (downtrend); a channel adds a parallel line; more touches = stronger line.

9 Moving averages (SMA, EMA) smooth prices; price above a rising MA is bullish; a golden cross is the 50-day crossing above the 200-day, a death cross the reverse.

10 RSI (typically 14-period) ranges 0–100; above 70 overbought, below 30 oversold. RSI = 100 - 100/(1 + RS), RS = average gain / average loss.

11 MACD = 12-period EMA - 26-period EMA; signal line = 9-period EMA of MACD; MACD crossing above signal is bullish.

12 Bollinger Bands: 20-period SMA ± 2 standard deviations; bands narrow in low volatility (squeeze) and widen in high volatility.

13 Volume should expand in the direction of the trend and on breakouts; divergences between price and an oscillator warn of weakening trends.

14 Candlestick signals: doji (indecision), hammer (bullish reversal after a fall), shooting star (bearish after a rise), bullish/bearish engulfing.

| | | |
|---|---|---|
| RSI overbought / oversold<br>**> 70 / < 30** | MACD<br>**EMA12 - EMA26; signal EMA9** | Bollinger Bands<br>**20 SMA ± 2 SD** |
| Golden cross<br>**50-DMA crosses above 200-DMA** | Dow primary trend<br>**≥ 1 year; secondary ~3 weeks-3 months** | |
| Dow phases<br>**Accumulation, public participation, distribution** | H&S target<br>**Head-to-neckline distance from breakout** | |

🎯 EXAM TIP

This chapter is 15 marks: master pattern classification (reversal vs continuation), Dow Theory tenets, indicator settings and readings, support/resistance role reversal, and simple target calculations.

⚠️ TRAP

Flags, pennants and most triangles are CONTINUATION patterns; head and shoulders and double tops are REVERSAL patterns. Also, a doji alone signals indecision, not a confirmed reversal.

## 🧮 Formulas

### Market capitalisation

```
Share price x Shares outstanding
```

₹450 x 20 crore shares = ₹9,000 crore

### Enterprise value

```
Market cap + Debt + Preference capital
+ Minority interest - Cash
```

₹9,000 cr + ₹2,000 cr debt - ₹500 cr cash = ₹10,500 crore

### EPS

```
PAT attributable to equity / Weighted
average shares
```

₹600 cr / 20 cr shares = ₹30

### P/E and earnings yield

```
P/E = Price / EPS; Earnings yield = EPS
/ Price
```

₹450 / ₹30 = 15×; earnings yield = 1/15 = 6.67%

### P/B

```
Price / Book value per share
```

Net worth ₹3,000 cr / 20 cr shares = BVPS ₹150; ₹450/₹150 = 3×

### PEG ratio

```
P/E ÷ expected EPS growth (%)
```

P/E 15, growth 20% → PEG 0.75

### EV/EBITDA → price

```
Target EV = multiple x EBITDA; Equity =
EV - Net debt; Price = Equity / Shares
```

10× x ₹1,200 cr = ₹12,000 cr; less net debt ₹1,500 cr = ₹10,500 cr; / 20 cr = ₹525

### Dividend yield

```
DPS / Price
```

₹9 / ₹450 = 2%

### Current yield (bond)

```
Annual coupon / Market price
```

8% coupon on ₹1,000 FV priced at ₹960 → 80/960 = 8.33%

### ROE

```
PAT / Average shareholders' equity
```

₹600 cr / ₹3,000 cr = 20%

### ROCE

```
EBIT / (Equity + Debt)
```

₹1,000 cr / (₹3,000 + ₹2,000) cr = 20%

### DuPont (3-step)

```
ROE = (PAT/Sales) x (Sales/Assets) x
(Assets/Equity)
```

10% x 1.2 x 1.5 = 18%

### Current and quick ratio

```
CA / CL; (CA - Inventory) / CL
```

CA ₹500, Inventory ₹200, CL ₹250 → 2.0 and 1.2

### Debt-equity and interest coverage

```
Debt / Equity; EBIT / Interest
```

₹2,000/₹3,000 = 0.67; ₹1,000/₹200 = 5×

### Working-capital days

```
Inventory days = Inv/COGS x 365; Receivable days = AR/Sales x 365; CCC = Inv days + AR days - AP days
```

60 + 45 - 30 = 75 days

### Free cash flow to firm

```
FCFF = EBIT(1 - t) + Depreciation - Capex - Increase in working capital
```

₹1,000 x 0.75 + 150 - 300 - 50 = ₹550 cr

### DCF / Gordon terminal value

```
TV = FCF(n+1) / (r - g); Value = Σ FCF/(1+r)^t + TV/(1+r)^n
```

FCFF 100, 110, 121; g 5%, WACC 12% → TV = 127.05/0.07 = 1,815; EV ≈ 263.1 + 1,291.9 = ₹1,555 cr

### Dividend discount model

```
P0 = D1 / (ke - g)
```

D1 ₹5, ke 12%, g 7% → ₹100

### CAPM

```
Ke = Rf + β (Rm - Rf)
```

7% + 1.2 x (13% - 7%) = 14.2%

### WACC

```
We x Ke + Wd x Kd x (1 - t)
```

0.6 x 14% + 0.4 x 10% x 0.75 = 8.4% + 3.0% = 11.4%

### Holding-period return

```
(P1 - P0 + Income) / P0
```

Buy ₹200, sell ₹230, dividend ₹4 → 34/200 = 17%

### CAGR

```
(End / Begin)^(1/n) - 1
```

₹100 → ₹161.05 in 5 years → 10%

### Sharpe ratio

```
(Rp - Rf) / σp
```

(15% - 7%) / 20% = 0.40

### Treynor ratio

```
(Rp - Rf) / βp
```

(15% - 7%) / 1.2 = 6.67

### Jensen's alpha

```
Rp - [Rf + βp(Rm - Rf)]
```

15% - [7% + 1.2 x 6%] = 0.8%

### Margin of safety

```
(Intrinsic value - Price) / Intrinsic value
```

IV ₹500, price ₹350 → 30%

### Theoretical ex-rights price

```
(N x P + R x S) / (N + R)
```

4 shares @ ₹200 + 1 right @ ₹150 → ₹190

### Bonus / split adjustment

```
Adjusted price = Old price x Old shares / New shares
```

1:1 bonus on ₹800 stock → ~₹400; split ₹10 FV to ₹2 on ₹1,000 → ~₹200

### Hedge ratio

```
h = ρ x σS / σF; Contracts = h x Exposure / Contract size
```

0.9 x 2%/2.5% = 0.72

### Cost of carry (futures fair value)

```
F = S x (1 + r x t) + storage costs (simple approximation)
```

S ₹5,000, 8% p.a., 3 months → ₹5,100 before storage

### RSI

```
100 - 100 / (1 + RS); RS = Avg gain / Avg loss
```

RS = 3 → RSI = 75 (overbought zone)

### Beta

```
Cov(Ri, Rm) / Var(Rm)
```

Cov 0.024, Var 0.02 → β = 1.2

## 🔢 Numbers to remember

**100 (80 MCQ + 5 cases x 4 Qs)**
Exam questions

**120 minutes / 60%**
Duration / pass mark

**25% of question marks**
Negative marking

**3 years**
Certificate validity

**20 January 2026**
Revised exam effective

**Technical Analysis - 15 marks**
Heaviest chapter

**Financial Analysis 12, Valuation 12, Legal & Regulatory 10**
Next heaviest

**75**
Part-time RA client cap

**₹1,51,000 per family per year**
Fee cap for individual/HUF clients

**Up to 1 year (client consent)**
Advance fee limit

**₹1L (≤150), ₹2L (151-300), ₹5L (301-1,000), ₹10L (>1,000 clients)**
Deposit tiers

**30 days before / 5 days after report**
Personal trading blackout

**40 days after pricing**
IPO quiet period (manager/co-manager)

**10 days after pricing**
FPO quiet period

**25 days from offering date**
IPO underwriter quiet period

**15 days before/after expiry**
Lock-up related quiet period

**1% or more**
Ownership disclosure threshold

**12 months**
Compensation disclosure look-back

**Rating ≥1 yr → period or 3 years, whichever shorter**
Price chart in report

**5 years**
Record retention

**21 calendar days**
Grievance resolution

**4% ± 2%**
RBI CPI target

**10% stake threshold**
FDI vs FPI

**T+1**
Equity settlement

**70 / 30 (14-period)**
RSI zones

**12, 26, 9**
MACD settings

**20-period, 2 SD**
Bollinger Bands

**50-DMA over 200-DMA**
Golden cross

**About 3 weeks to 3 months**
Dow secondary trend

**5 forces / 6 factors / 4 quadrants**
Porter / PESTLE / BCG

**8**
RA Code of Conduct heads

**₹2,000 application + ₹3,000 registration (every 5 years)**
SEBI registration fee (individual RA)

## 📖 Glossary

| | |
|---|---|
| **Research analyst** | A person who, for consideration, provides research services - reports, buy/sell/hold calls, price targets, model portfolios or trading calls - on listed or to-be-listed securities. |
| **Research entity** | An intermediary (broker, merchant banker, investment banker, underwriter) that also issues research reports in its own name through analysts it employs. |
| **Part-time research analyst** | An individual or partnership firm that provides research services while also engaged in another business or employment; capped at 75 clients. |
| **Subject company** | The company whose securities are covered by a research report or a public appearance. |
| **Public appearance** | Speaking in seminars, TV, conference calls, print or social media where the analyst gives a recommendation or opinion on securities. |
| **RAASB** | Research Analyst Administration and Supervisory Body - BSE Ltd, recognised by SEBI; RAs enlist with it and their deposits are lienied to it. |
| **UPSI** | Unpublished price-sensitive information - non-public information that would likely materially affect the share price if published. |
| **Mosaic theory** | Reaching a conclusion by combining public information and non-material non-public details; legitimate research practice. |
| **Front running** | Trading ahead of a known pending order or research publication to profit from its price impact; prohibited. |
| **E-I-C analysis** | Economy-Industry-Company framework of top-down fundamental analysis. |
| **Enterprise value** | Value of the entire business to all capital providers: market cap plus debt and other claims, minus cash. |
| **EBITDA** | Earnings before interest, tax, depreciation and amortisation - a proxy for operating cash profit. |
| **ROCE** | Return on capital employed = EBIT / (equity + debt); measures operating efficiency of all capital. |
| **DuPont analysis** | Splitting ROE into net margin, asset turnover and financial leverage to see what drives returns. |
| **Free cash flow** | Cash left after operating expenses, taxes, working capital and capex; FCFF belongs to all investors, FCFE to equity holders. |
| **Terminal value** | Value of all cash flows beyond the explicit forecast period, often via the Gordon growth formula. |
| **SOTP** | Sum-of-the-parts: valuing each business segment separately and adding them up, often with a holding-company discount. |

| | |
|---|---|
| **Transaction multiple** | Valuation multiple paid in a past M&A deal; usually higher than trading multiples because it includes a control premium. |
| **PEG ratio** | P/E divided by expected earnings growth (%); below 1 is often read as cheap relative to growth. |
| **Beta** | Sensitivity of a stock's return to market returns; a measure of systematic risk. |
| **Contango** | Futures price above spot price, typically reflecting cost of carry. |
| **Backwardation** | Futures price below spot price, often signalling tight current supply. |
| **Hedge ratio** | Proportion of an exposure covered with futures; minimum-variance ratio = correlation x σspot/σfutures. |
| **TERP** | Theoretical ex-rights price - the expected share price after a rights issue. |
| **Demerger** | Splitting a business division into a separate company whose shares are given to existing shareholders. |
| **ASM** | Additional Surveillance Measure - exchange action (e.g., higher margins) on stocks with unusual price/volume behaviour. |
| **GSM** | Graded Surveillance Measure - stage-wise trading restrictions on stocks whose price is not commensurate with financial health. |
| **Dow Theory** | Charles Dow's framework of primary, secondary and minor trends, confirmation by averages and volume. |
| **Head and shoulders** | A three-peak reversal pattern where the middle peak (head) is highest; a neckline break confirms a downtrend. |
| **RSI** | Relative Strength Index - momentum oscillator from 0 to 100; above 70 overbought, below 30 oversold. |
| **MACD** | Moving Average Convergence Divergence - difference of 12– and 26–period EMAs with a 9–period signal line. |
| **Margin of safety** | The discount of price to estimated intrinsic value that protects against errors in analysis. |
| **Accredited investor** | An investor certified as financially sophisticated; RA fee caps for individuals/HUFs do not apply to them. |

## ⚡ Quick revision: the night before

1. Exam: 80 MCQs + 5 cases x 4 = 100 questions, 100 marks, 2 hours, pass 60, minus 0.25 per wrong answer.
2. Technical Analysis = 15 marks; Financial Analysis and Valuation = 12 each; Legal & Regulatory = 10.
3. EV = market cap + debt + preference + minority - cash.
4. Bond price and yield move inversely; zero-coupon duration = maturity.
5. Contango: futures > spot. Backwardation: futures < spot. Basis = spot - futures.
6. Top-down = Economy → Industry → Company; bottom-up starts with the company.
7. Fiscal policy = government; monetary policy = RBI. RBI CPI target 4% ± 2%.
8. FDI ≥ 10% stake; FPI < 10%.
9. Weaker rupee helps exporters (IT, pharma), hurts importers (OMCs).
10. Porter: entrants, suppliers, buyers, substitutes, rivalry. PESTLE has 6 letters.
11. BCG: Star (high growth, high share), Cash Cow (low growth, high share), Question Mark (high growth, low share), Dog (low, low).
12. SWOT: S and W internal; O and T external.
13. ROE = PAT/Equity; ROCE = EBIT/(Equity + Debt).
14. DuPont ROE = net margin x asset turnover x equity multiplier.
15. Quick ratio excludes inventory. CCC = inventory days + receivable days - payable days.
16. Profits up but CFO down = earnings-quality red flag.
17. Bonus: reserves → share capital, face value same. Split: face value falls, reserves same.
18. TERP = (N x P + R x S) / (N + R).
19. Buyback cuts share count → EPS rises.
20. FCFF + WACC → EV; FCFE + Ke → equity value.
21. Gordon: D1/(ke - g); needs g < ke.
22. CAPM: Ke = Rf + β(Rm - Rf).
23. PEG = P/E ÷ growth %. Earnings yield = 1/(P/E).
24. Transaction multiples > trading multiples because of control premium.
25. Producer hedges with short futures; consumer with long futures. Hedge ratio = ρ σS/σF.
26. Strong dollar usually pulls dollar commodity prices down.
27. Sharpe uses σ (total risk); Treynor uses β; Jensen's alpha = Rp - CAPM return.
28. Systematic risk can't be diversified; unsystematic can.
29. Report rating must define meaning, horizon and benchmark; price chart for period or 3 years (whichever shorter) if rating held ≥1 year.
30. No RA registration needed for IAs, CRAs, AMCs issuing research - but conduct rules still apply.
31. Part-time RA: max 75 clients; must use the term 'part-time research analyst'.
32. Fee cap: ₹1,51,000 per family per year for individual/HUF clients; advance up to 1 year with consent.
33. Deposit: ₹1L / ₹2L / ₹5L / ₹10L by client tiers 150 / 300 / 1,000 / above; RAASB = BSE.
34. Blackout: 30 days before and 5 days after report; never trade against your own call.
35. Quiet periods: 40 days after IPO pricing, 10 days after FPO pricing (manager/co-manager).
36. Disclose 1%+ holding, 12-month compensation, directorships, market-making, AI use.
37. Records 5 years; grievance within 21 calendar days; annual compliance audit.
38. ASM = unusual price/volume; GSM = price not justified by fundamentals.
39. Dow: primary ≥1 yr, secondary ~3 weeks-3 months, minor <3 weeks; volume confirms.
40. Reversal: H&S, double/triple top-bottom, rounding. Continuation: flags, pennants, triangles, rectangles.
41. RSI > 70 overbought, < 30 oversold; MACD 12-26-9; Bollinger 20 SMA ± 2 SD; golden cross 50 over 200.

## 🗓️ 21-day study plan (2 h/day)

**Days 1–3: Foundations: Chapters 1–5 (profession, market, terminology, research approaches, economy)**

- Read notes for Ch 1–5 and make a one-page cheat sheet of terms
- Practise EV, market cap and current-yield sums
- Take Mock 1 untimed and review every explanation

**Days 4–7: Industry and company: Chapters 6–8**

- Memorise Porter, PESTLE, BCG, SCP with one example each
- Work 20 ratio problems (ROE, ROCE, DuPont, CCC, liquidity)
- Take Mock 2 and log weak areas

**Days 8–11: Corporate actions and valuation: Chapters 9–10**

- Drill TERP, bonus/split, buyback and swap-ratio sums
- Do 10 DCF/DDM/CAPM/EV-EBITDA problems with a calculator
- Take Mock 3

**Days 12–14: Commodities, risk-return and report quality: Chapters 11–13**

- Hedge ratio and contango/backwardation drills
- Sharpe/Treynor/Jensen/CAGR numericals; list the 8 biases with an example each
- Take Mock 4

**Days 15–18: Regulation and technical analysis: Chapters 14–15 (25 marks together)**

- Make flashcards of every RA Regulation number (75, ₹1.51L, 30/5, 40/10, 1%, 12 months, 5 yrs, 21 days)
- Sketch every chart pattern and label reversal vs continuation; memorise indicator settings
- Take Mock 5

**Days 19–21: Full-length practice and revision**

- Take Mock 6 and Mock 7 timed; analyse every wrong answer
- Run the 100-question simulator under exam conditions
- Night before: Quick Revision list and Numbers to Remember only

Independent study notes prepared from the official NISM workbook. Verify against the latest workbook on nism.ac.in.